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PowerStake:
Advanced Engineering Design

EGH419: Advanced Design and Entrepreneurship

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1 Preliminary Design – Web Interface

The PowerStake web interface acts as the primary interaction point between the user and the system. It is responsible for presenting appliance-level power monitoring data, AI-generated recommendations, system alerts, and manual control options in a clear and accessible format. The interface supports the validation plans developed in Assessment 2 by allowing users to view system outputs and respond to recommendations during prototype testing.

The interface was implemented using a lightweight web-based dashboard approach, with Streamlit selected for the prototype. Streamlit was chosen because it allows rapid dashboard development, integrates directly with Python-based AI logic, and is suitable for low TRL systems where functional validation is more important than production-level frontend development. This is appropriate for the current TRL 3–4 stage, where the purpose of the interface is to demonstrate system functionality and validate user interaction rather than deliver a fully commercial application.

The core features of the interface include:

- real-time display of appliance power usage data;
- alerts for abnormal appliance behaviour;
- AI-generated energy-saving recommendations;
- device-level monitoring overview; and
- manual override controls for switching actions.

The system architecture for the interface follows a simple data flow:

Power Monitoring → AI Subsystem → Web Interface → User Decision → System Control

This ensures that data collected from the hardware and processed by the AI subsystem is translated into actionable outputs for the user. The web interface therefore provides the link between the technical monitoring system and the customer-facing value proposition of PowerStake.

From a feasibility perspective, the web interface is achievable within the project timeframe because it avoids the need for complex frontend and backend separation. The use of Streamlit allows the dashboard to be developed alongside the AI subsystem using the same Python environment. However, limitations include reduced scalability, limited customisation, and the absence of production-level user authentication. These limitations are acceptable for a prototype-level implementation but would need to be addressed before commercial release.

The PowerStake web interface prototype is available at:

`https://egh419assessment3.streamlit.app/`

The interface design was developed to satisfy the key user-facing requirements of the PowerStake prototype: displaying live appliance-level power data, presenting AI-generated recommendations, alerting users to abnormal appliance behaviour, and allowing manual override of suggested control actions. The main design constraints were development time, low prototype cost, integration with Python-based analysis tools, and the need for a simple interface that could be tested without a full mobile application backend.

Screenshots of the implemented interface are provided below to demonstrate the dashboard layout, system outputs, and user interaction features.

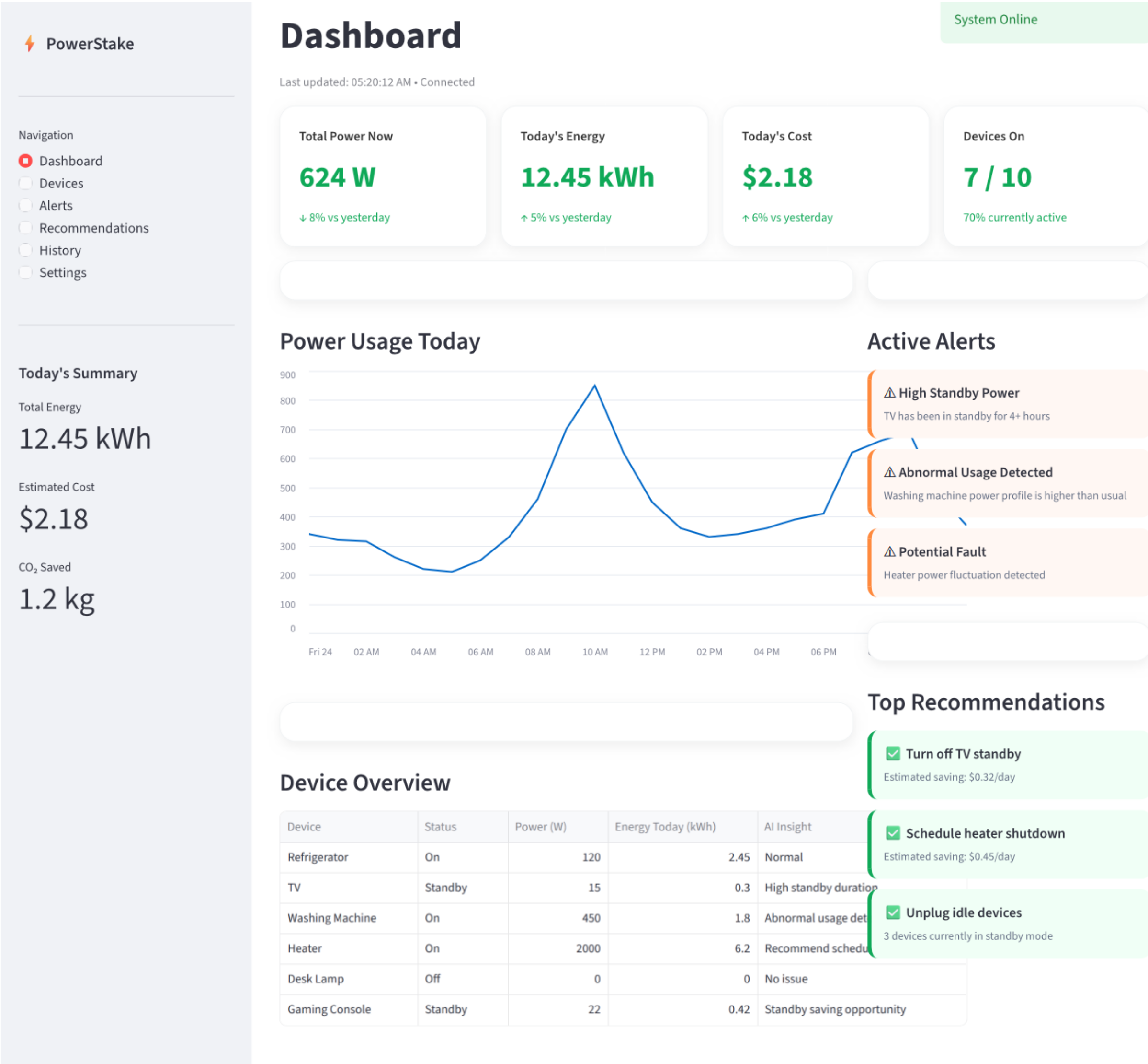

Figure 1: PowerStake web interface dashboard overview

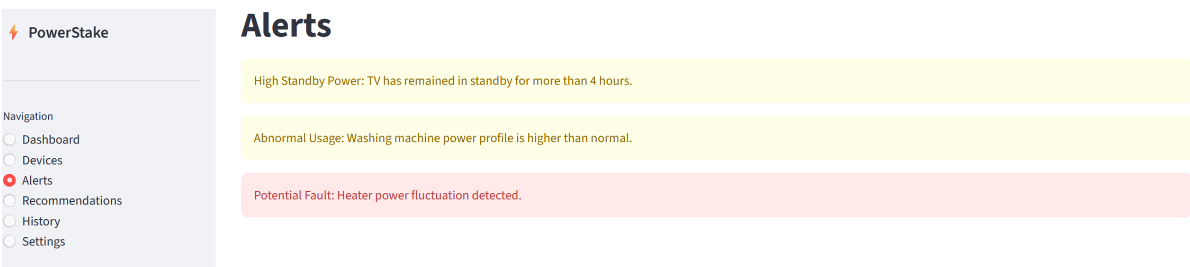

Figure 2: PowerStake web interface system alert display

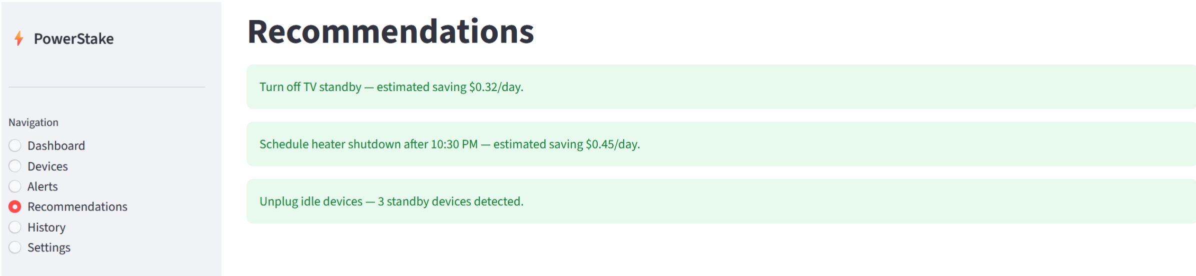

Figure 3: PowerStake web interface recommendation and interaction features

2 Intellectual Property (IP)

A preliminary IP search was conducted using Google Patents, IP Australia, and competitor product research. Existing smart plug and energy-monitoring products, such as TP-Link and Belkin devices, already provide remote switching, scheduling, mobile control, and energy monitoring functions [1], [2]. Related patents also show that device-state detection using smart plug power data and home energy monitoring/control systems are already protected [3], [4], [5]. Therefore, PowerStake should not attempt to protect the broad concept of a smart plug or connected power board.

For this preliminary web interface design, the main asset requiring protection is the PowerStake dashboard and recommendation workflow. This includes the Streamlit dashboard source code, screen layout, alert cards, recommendation panels, and the way AI-generated outputs are displayed to users. The source code is automatically protected by copyright once created, as software code is treated as copyright-protected material in Australia [6], [7]. The visual layout may also be protected through design rights if the interface becomes commercially distinctive. IP Australia states that registering a design costs a minimum of $200 [8]. The PowerStake name and logo should also be protected through trade mark registration before public release or crowdfunding, with standard applications starting from $250 per class using IP Australia's picklist system [9].

Trade secrets are also suitable for protecting unpublished recommendation ranking rules and interface logic, as these can be kept confidential through private repositories, restricted access, and non-disclosure agreements.

Table 1: Recommended IP protection strategy for the PowerStake web interface

Asset	**Protection Type**	**Estimated Cost**	**Justification**
Dashboard source code	Copyright	Automatic / low cost	Protects the original web interface implementation.
Dashboard layout and screen design	Design rights	From $200	Protects the visual appearance of the interface.
PowerStake name and logo	Trade mark	From $250/class	Protects brand identity before launch.
Recommendation display logic	Trade secret	Low internal cost	Protects confidential user-facing decision logic.

3 Risk Assessment

The PowerStake web interface creates risks related to user decision-making and the reliability of displayed information. A relevant regulation is the Australian Privacy Act 1988, including the Australian Privacy Principles, as household energy data may reveal user behaviour patterns such as occupancy, appliance use, and daily routines [10]. The interface should also consider WCAG 2.2 accessibility guidance so that alerts, charts, and recommendations remain readable and usable for different users [11].

		Consequences				
		Insignificant (1) No injuries / minimal financial loss	**Minor (2)** First aid treatment / medium financial loss	**Moderate (3)** Medical treatment / high financial loss	**Major (4)** Hospitable / large financial loss	**Catastrophic (5)** Death / massive financial loss
Likelihood	**Almost Certain (5)** Often occurs / once a week	Moderate (5)	High (10)	High (15)	Catastrophic (20)	Catastrophic (25)
	Likely (4) Could easily happen / once a month	Moderate (4)	Moderate (8)	High (12)	Catastrophic (16)	Catastrophic (20)
	Possible (3) Could happen or known it to happen / once a year	Low (3)	Moderate (6)	Moderate (9)	High (12)	High (15)
	Unlikely (2) Hasn't happened yet but could / once every 10 years	Low (2)	Moderate (4)	Moderate (6)	Moderate (8)	High (10)
	Rare (1) Conceivable but only on extreme circumstances / once in 100 years	Low (1)	Low (2)	Low (3)	Moderate (4)	Moderate (5)

Figure 4: Risk assessment matrix

The risk assessment matrix (Figure 4) was applied by identifying the nature, cause, risk event, effect, likelihood, impact, mitigation, and contingency response for each risk [12].

Table 2: Web interface risk assessment

Risk Item	**R1: Poor dashboard usability**	**R2: Delayed or outdated displayed data**
Risk Type	Operational / customer adoption risk	Technical / operational feasibility risk
Nature	The user may not understand or act on the dashboard outputs.	The web interface may display information that is not current.
Cause	Too many alerts, unclear recommendation wording, poor layout, or confusing energy data visualisation.	Slow data refresh, unstable connection, cached values, or delayed communication between the AI subsystem and interface.
Risk Event	Users ignore recommendations or misunderstand what action should be taken.	The dashboard shows outdated power data, device status, alerts, or recommendations.
Effect	Reduced customer value, weaker prototype validation, and lower user trust in PowerStake.	Incorrect user decisions, reduced trust, and unreliable validation results.
Likelihood	Possible (3)	Possible (3)
Impact	Major (4)	Moderate (3)
Risk Score	12 – High	9 – Moderate
Mitigation	Use clear visual hierarchy, simple wording, limited high-priority alerts, and user testing during prototype validation.	Include last-updated timestamps, connection status, automatic refresh, and stale-data warnings.
Contingency	Redesign the dashboard around fewer key outputs and retest with users before further validation.	Disable automated recommendations, show a warning message, and verify backend data before continuing testing.

These risks are most relevant to the TRL 3–4 prototype stage because the web interface must demonstrate that PowerStake outputs are understandable, timely, and useful before further technical development or household pilot testing. Managing these risks improves both operational feasibility and the reliability of prototype validation.

References

[1] TP-Link Technologies. "Smart plug product range." Competitor smart plug products with mobile control, scheduling and monitoring features, Accessed: May 29, 2026. [Online]. Available: `https://www.tp-link.com/au/home-networking/smart-plug/`.

[2] Belkin International. "Wemo smart plugs and smart home products." Competitor smart-home plug and connected device products, Accessed: May 29, 2026. [Online]. Available: `https://www.belkin.com/products/wemo/`.

[3] Google Patents. "Us11182699b2: Determining device state changes using smart plugs." Patent relating to determining device state changes using smart plug power-monitoring signals, Accessed: May 29, 2026. [Online]. Available: `https://patents.google.com/patent/US11182699/en`.

[4] Google Patents. "Us8996188b2: System and method for home energy monitor and control." Patent relating to household energy monitoring and control, Accessed: May 29, 2026. [Online]. Available: https://patents.google.com/patent/US8996188B2/en.

[5] Google Patents. "Us10750252b2: Identifying device state changes using power data and network data." Patent relating to identifying device state changes using power data and network data, Accessed: May 29, 2026. [Online]. Available: https://patents.google.com/patent/US10750252B2/en.

[6] IP Australia. "What computer-related inventions can be patented." Explains that source code and executable code are automatically protected by copyright in Australia, Accessed: May 29, 2026. [Online]. Available: https://www.ipaustralia.gov.au/patents/what-are-patents/what-computer-related-inventions-can-be-patented.

[7] Business Victoria. "4 intellectual property considerations for software ownership." Explains that newly written software code is automatically protected by copyright, Accessed: May 29, 2026. [Online]. Available: https://business.vic.gov.au/learning-and-advice/hub/4-intellectual-property-considerations-for-software-ownership.

[8] IP Australia. "Timeframes and fees: Design rights." States that registering a design in Australia costs a minimum of 200 Australian dollars, Accessed: May 29, 2026. [Online]. Available: https://www.ipaustralia.gov.au/design-rights/timeframes-and-fees.

[9] IP Australia. "Trade mark timeframes and fees." States that a standard trade mark application with picklist costs 250 Australian dollars per class, Accessed: May 29, 2026. [Online]. Available: https://www.ipaustralia.gov.au/trade-marks/timeframes-and-fees.

[10] Office of the Australian Information Commissioner. "Australian privacy principles." Australian privacy framework under the Privacy Act 1988 relevant to handling personal information, Accessed: May 29, 2026. [Online]. Available: https://www.oaic.gov.au/privacy/australian-privacy-principles.

[11] World Wide Web Consortium. "Web content accessibility guidelines 2.2." Accessibility standard for designing readable and usable web interfaces, Accessed: May 29, 2026. [Online]. Available: https://www.w3.org/TR/WCAG22/.

[12] Queensland University of Technology, *Egh419 week 9 tutorial: Ip landscape and strategy and risk assessment*, Course workshop material outlining risk assessment method, likelihood-impact matrix, and risk management strategies, 2025.

3.1 Generative AI

Generative AI was used as a support tool during the preparation of this report. It was used to assist with grammar, clarity, structure, and checking alignment with the assessment requirements. Final decisions, technical content, design choices, references, and report editing were completed by the author.

General Prompts

- "Can you check this writing for readability, sentence structure, and consistency while keeping my original technical meaning?"
- "Can you help tighten this paragraph so it reads more clearly in an engineering report style?"

Preliminary Design

- "Can you review my web interface section and tell me whether it clearly covers what the prototype does, why it was built this way, and what its limitations are?"
- "Can you refine my explanation of the Streamlit dashboard so it sounds clearer but still reflects my own design choices?"

Intellectual Property

- "Can you help shorten my competitor IP research so it stays focused on the dashboard and user interface rather than the full smart plug product?"
- "Can you convert the IP and product sources I used into BibTeX entries for my references file?"

Risk Assessment

- "Can you suggest web-interface-related risks that are separate from my teammate's software and safety risks?"
- "Can you check whether my likelihood and impact scores make sense for dashboard usability and stale-data risks?"